# Mass Gap for the O($N$) Model at Large $N$ via Brascamp-Lieb and Feynman-Kac

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**Abstract**

We outline a proof of the mass gap for the O($N$) linear sigma model in two Euclidean dimensions at large $N$. The proof has three steps: (1) the gap equation determines the saddle point $\sigma_*$ and the physical mass $m_0 = \sqrt{\sigma_*}$ (dimensional transmutation), (2) Brascamp-Lieb gives $\sigma$-concentration at $\sigma_*$ with $O(1/N)$ variance, and (3) Feynman-Kac controls the $1/N$ corrections to the propagator mass. The convexity condition for Brascamp-Lieb, $\sigma_* > 1/(16\pi\lambda)$, is a condition on the renormalized mass vs. the coupling and is cutoff-independent.

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# 1 Setup

## 1.1 The O($N$) linear sigma model

Let $\varphi = (\varphi^1, \ldots, \varphi^N) : \Lambda \to \mathbb{R}^N$ on a lattice $\Lambda \subset \mathbb{Z}^2$. The action is

$$S[\varphi] = \tfrac{1}{2} \sum_i \langle \varphi^i, (-\Delta)\varphi^i \rangle + N \sum_x \lambda\big(\tfrac{|\varphi(x)|^2}{N} - v^2\big)^2. \tag{1}$$

The bare parameter $v^2$ is adjusted as a function of the UV cutoff $\Lambda$ to keep the physical mass fixed (the only renormalization needed in 2d; the quartic coupling $\lambda$ does not run).

## 1.2 Hubbard-Stratonovich

Setting $\sigma(x) = |\varphi(x)|^2/N$ and integrating out $\varphi$:

$$\langle \varphi^i(x)\varphi^i(0)\rangle = \mathbb{E}_\sigma\big[(-\Delta + \sigma)^{-1}(x, 0)\big], \tag{2}$$

where $\mu_\sigma \propto e^{-N s_{\text{eff}}[\sigma]}$ with

$$s_{\text{eff}}[\sigma] = \tfrac{1}{2}\mathrm{Tr}\log(-\Delta + \sigma) + \sum_x \lambda(\sigma(x) - v^2)^2. \tag{3}$$

All $N$ components of $\varphi$ see the same potential $\sigma$. There is no distinction between radial and angular modes in the conditional measure.

# 2 The gap equation

## 2.1 Saddle point of $s_{\text{eff}}$

For uniform $\sigma(x) = \sigma_0$, the stationarity condition $ds_{\text{eff}}/d\sigma_0 = 0$ is

$$\tfrac{1}{2}G(x, x; \sigma_*) = 2\lambda(v^2 - \sigma_*), \tag{4}$$

where $G(x, x; \sigma_0) = \int \frac{d^2k}{(2\pi)^2} \frac{1}{k^2+\sigma_0}$. Note that $v^2$ is *not* a stationary point: at $\sigma_0 = v^2$ the derivative $\frac{1}{2}G(x, x; v^2) > 0$, so the minimum lies at $\sigma_* < v^2$.

In 2d with cutoff $\Lambda$: $G(x, x; \sigma_*) \approx \frac{1}{4\pi} \log \frac{\Lambda^2}{\sigma_*}$, giving

$$\frac{1}{8\pi} \log \frac{\Lambda^2}{\sigma_*} = 2\lambda(v^2 - \sigma_*). \tag{5}$$

For $\sigma_* \ll v^2$: $\sigma_* \approx \Lambda^2 e^{-16\pi\lambda v^2}$, exponentially small (dimensional transmutation).

### 2.2 Physical mass and renormalization

The physical mass at $N = \infty$ is $m_0 = \sqrt{\sigma_*}$, since $\sigma = \sigma_*$ deterministically and $(-\Delta + \sigma_*)^{-1}(x, 0) \sim e^{-m_0|x|}$.

In the continuum limit ($\Lambda \to \infty$), we adjust the bare parameter:

$$v^2(\Lambda) = \sigma_* + \frac{1}{16\pi\lambda} \log \frac{\Lambda^2}{\sigma_*} \tag{6}$$

to keep $\sigma_*$ (and hence $m_0$) fixed. The quartic coupling $\lambda$ does not run (super-renormalizable in 2d).

## 3 Brascamp-Lieb concentration

### 3.1 Hessian at $\sigma_*$

The Hessian of $s_{\text{eff}}$ at $\sigma_*$, in Fourier, is

$$H(k) = 2\lambda - \tfrac{1}{2}\Pi(k), \quad \Pi(k) = \int \frac{d^2p}{(2\pi)^2} \frac{1}{(p^2 + \sigma_*)((k-p)^2 + \sigma_*)}. \tag{7}$$

At $k = 0$: $\Pi(0) = 1/(4\pi\sigma_*)$, so $H(0) = 2\lambda - 1/(8\pi\sigma_*)$.

### 3.2 Convexity condition

$H(k)$ is positive definite for all $k$ when $H(0) > 0$, i.e.,

$$\boxed{\sigma_* > \frac{1}{16\pi\lambda}.} \tag{8}$$

This is a condition on the **renormalized** parameters $\sigma_*$ and $\lambda$. It is **cutoff-independent**: since $\lambda$ does not run and $\sigma_*$ is held fixed by renormalization, the condition (10) either holds or fails independently of $\Lambda$. In particular, it survives the continuum limit.

Equivalently: the physical mass must satisfy $m_0 > 1/\sqrt{16\pi\lambda}$. For $\lambda = 1$: $m_0 > 0.14$.

### 3.3 Brascamp-Lieb bounds

When (10) holds, $s_{\text{eff}}$ is strictly convex at $\sigma_*$ with convexity parameter $\kappa = 2\lambda - 1/(8\pi\sigma_*) > 0$. The measure $e^{-Ns_{\text{eff}}}$ is log-concave, and Brascamp-Lieb [1] gives:

$$\text{Cov}(\sigma(x), \sigma(y)) \leq \frac{1}{N} H^{-1}_{xy}, \tag{9}$$

$$\text{Var}(\sigma(x)) \leq \frac{1}{\kappa N}, \tag{10}$$

$$|\kappa_k(\sigma(x))| \leq \frac{C^k\, k!}{(\kappa N)^{k-1}}, \quad k \geq 2. \tag{11}$$

The $\sigma$-field concentrates at $\sigma_*$ with $O(1/\sqrt{N})$ fluctuations.

# 4 Feynman-Kac and $1/N$ corrections

## 4.1 Leading order ($N = \infty$)

At $N = \infty$: $\sigma = \sigma_*$ deterministically, the gap equation determines $\sigma_*$, and the mass is $m_0 = \sqrt{\sigma_*}$. No further analysis needed.

## 4.2 Finite $N$: cumulant expansion

From (2) and the Feynman-Kac formula, the $\varphi$-propagator involves $\mathbb{E}_\sigma[e^{-\int \sigma(\omega)\,dt}]$. The cumulant expansion:

$$\mathbb{E}_\sigma[e^X] = \exp\Big(\sum_{k=1}^{\infty} \frac{\kappa_k(X)}{k!}\Big), \quad X = -\int_0^s \sigma(\omega(t))\,dt. \tag{12}$$

- $\kappa_1 = -s\sigma_*$ (leading mass, from the gap equation).
- $\kappa_2 = O(s/(\kappa N))$ (from BL variance bound).
- $\kappa_k = O(s/(\kappa N)^{k-1})$ for $k \geq 3$ (from BL cumulant bounds).

The corrected mass:

$$m^2 = \sigma_*\big(1 + O(1/(\kappa N))\big). \tag{13}$$

For $N \gg 1/\kappa$: the correction is small and $m^2 > 0$.

## 4.3 Main theorem

**Theorem 1** (Mass gap at large $N$)**.** *For the O(N) LSM on $\mathbb{Z}^2$ with $N \geq 3$ and $\sigma_* > 1/(16\pi\lambda)$ (convexity condition), there exists $N_0 = O(1/\kappa)$ such that for $N \geq N_0$, the connected two-point function decays exponentially:*

$$|\langle \varphi^i(x)\varphi^i(0)\rangle_c| \leq C\, e^{-m|x|} \tag{14}$$

*with $m = \sqrt{\sigma_*}\,(1 + O(1/(\kappa N))) > 0$, where $\sigma_*$ is the gap equation solution* (5)*.*

*Proof outline.*

1. **Gap equation** (§2): $\sigma_*$ is the minimum of $s_{\text{eff}}$, with $m_0 = \sqrt{\sigma_*} > 0$.
2. **Brascamp-Lieb** (§3): At $\sigma_*$, the Hessian is positive (by (10)), giving $\sigma$-concentration with $\text{Var} = O(1/(\kappa N))$ and cumulant bounds.
3. **Feynman-Kac** (§4): The cumulant expansion gives $m^2 = \sigma_*(1 + O(1/(\kappa N)))$. For $N > C/\kappa$: the correction is $< 1$ and $m > 0$.
4. **Exponential decay**: Standard from Feynman-Kac with $m > 0$. □

# 5 Discussion

## 5.1 The convexity condition is physical

The condition $\sigma_* > 1/(16\pi\lambda)$ is equivalent to $m_0^2 > 1/(16\pi\lambda)$, a lower bound on the physical mass. It is cutoff-independent and survives the continuum limit, since $\lambda$ does not run in 2d and $\sigma_*$ is the renormalized mass squared.

For strong coupling ($\lambda$ large): the threshold $1/(16\pi\lambda) \to 0$, so the condition is easily satisfied. For any fixed $\lambda > 0$: the condition holds for $m_0 > 1/\sqrt{16\pi\lambda}$.

## 5.2 No cluster expansion needed

Unlike Kupiainen [2], who uses a cluster expansion to control the $1/N$ corrections for the NLSM, our proof uses only:

- The gap equation (a single self-consistent equation),
- Brascamp-Lieb concentration (a convexity-based inequality),
- Feynman-Kac cumulant expansion (path integral representation).

The key simplification: the convexity of $s_{\text{eff}}$ at $\sigma_*$ (when (10) holds) makes the $\sigma$-measure log-concave, replacing the cluster expansion with BL.

## 5.3 The $N = 2$ obstruction

For $N = 2$: the BKT transition occurs at finite $\lambda$ (Dario-Garban [4]), driven by vortices ($\pi_1(S^1) = \mathbb{Z}$). The $\sigma$-field cannot detect angular topology. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no BKT.

## 5.4 Comparison with Kupiainen

| | This work | Kupiainen [2] |
|---|---|---|
| Model | LSM (finite $\lambda$) | NLSM ($\lambda = \infty$) |
| $\sigma$-control | Brascamp-Lieb | Cluster expansion |
| $\varphi$-decay | Feynman-Kac | Random walk expansion |
| Mass formula | $m = \sqrt{\sigma_*}(1 + O(1/N))$ | $m = m_0(1 + O(1/N))$ |
| Continuum limit | Yes (renormalized) | Yes |

Since the LSM and NLSM share the same infrared physics (they differ only at scale $M_\sigma \sim \sqrt{\lambda}\, v$), both proofs establish the same physical mass gap in the large-$N$ limit.

# References


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[3] O. A. McBryan and T. Spencer, Comm. Math. Phys. **53** (1977) 299–302.

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